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ENHANCED FILM COOLING EFFECTIVENESS WITH NEW SHAPED HOLES

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ABSTRACT

Gas Turbine Engines operate at temperatures higher than current material temperature limits. This necessitates cooling the metal through internal or external means and/ or protecting the metal with coatings that have higher material limits. Film cooling is one of the major technologies allowing today's gas turbines to operate at extremely high turbine inlet temperatures, consequently higher power density, and extend the cooled components life. Film cooling is a technique where a coolant is blown over the surface exposed to hot gas and a film of low temperature gas is maintained that protects the metal surface from the hot gas. The application of effective film-cooling techniques provides the first and best line of defense for hot gas path surfaces against the onslaught of extreme heat fluxes, serving to directly reduce the incident convective heat flux on the surface.

The effectiveness of film cooling methods depends on the blowing ratio, shape of the cooling holes, and geometrical parameters such as the area ratio and diffusion angle. Film cooling is performed almost exclusively through the use of discrete holes. The holes can be of round or other shaped. A detailed study of the literature shows that the fan shaped has higher effectiveness when compared to other shapes.

In this study a number of cooling hole shapes are evaluated numerically using the Computational Fluid Dynamics (CFD) tool ANSYS-CFX-11.0 with the objective of improving cooling effectiveness under a favorable pressure gradient main flow. In order to delineate the effects of shape from that of diffusion, a constant area ratio is assumed first. In the next set of analyses the effect of hole exit diffusion is considered. Results are presented in terms of surface temperatures and adiabatic effectiveness at three different blowing ratios for the different film cooling hole shapes analyzed. Comparison is made with reference to the fan shaped film cooling hole with forward and lateral angles of 10/10/10 degree respectively. Hole shapes that show improvement over the fan shaped hole are identified and optimized.

NOMENCLATURE

DR	density ratio
D	film hole diameter
M	blowing ratio $(\rho V)_{coolant}/(\rho V)_{main}$
VR	velocity ratio $V_{coolant}/V_{main}$
V	velocity (m/s)
μ	viscosity of air (kg/m s)
ρ	density of air (kg/m^3)
η	Effectiveness
T_m	Mainstream flow Total temperature
T_c	Coolant flow Total temperature

INTRODUCTION

In order to achieve higher efficiency and lower fuel consumption and emissions, modern gas turbines operate at higher temperatures. To allow this high temperature operation, turbine designers employ a variety of technologies. Film cooling is one such technology which designers rely upon. Film cooling provides relatively low fluid temperature at the surface of the airfoil through the use of discrete holes primarily. This provides a protective layer from the gas path temperatures. The goodness measure of a film cooling arrangement is determined primarily by its film cooling effectiveness.

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State-of-the-art film cooling schemes with shaped holes of various geometry are employed in advanced high-pressure turbines. To achieve maximum benefit from the expended cooling air, film cooling configurations have to be very accurately adjusted to the local thermal and aerodynamic conditions. High-performance propulsion gas turbine engines require cooling to protect the airfoil from thermal stress fields created by exposure to hot combustion gases. In the airfoil cooling process, compressor bleed air is introduced into the hollow core of each airfoil and is channeled and then dumped through the airfoil surface via rows of holes located in the vicinity of the leading edge of the airfoil and at other high thermally loaded locations along the suction and pressure surfaces. The art and science of film cooling concerns the bleeding of internal component cooling air through the external walls to form a protective layer of cooling between the hot gases and the component external surfaces. Because of its high importance and widespread application, research into the many aspects of film cooling has seen a tremendous increase in the last 10–15 years

Film cooling has been studied by many investigators for more than 30 years to understand the fundamental physics of film cooling and to improve the state of the art. The studies include injection through a single hole and multiple hole arrangements, as well as the effects of geometry. Goldstein provided an early review of these film cooling studies. Goldstein et al. [1] studied the effect of single and row of holes on the film cooling effectiveness. Study also includes the effect of lateral injection on the cooling performance. Goldstein [2] provided analytical approach to calculate the effectiveness of two dimensional slot injection. This study is based on the heat sink model and energy balance on the turbulent boundary layer. Wilson and Goldstein [3] carried out an experimental and theoretical investigation to examine the effect of film cooling injection from a two-dimensional slot on the downstream surface heat transfer for a high speed flow (Mach 3.0 mainstream). Goldstein et al. [4] studied the effect of hole geometry and density on three dimensional film cooling and found that the widened coolant passages at the exit improves the cooling effectiveness significantly. Eriksen and Goldstein [5] carried out a study on the effect of inclined circular tubes on the heat transfer and film cooling effectiveness. Effect of Slot Injection on the adiabatic wall effectiveness of a turbulent boundary layer was studied by Mayle and Kopper [6]. Flow visualization study on the Discrete Hole film cooling was carried out by Colladay and Russell [7]. Petersen et al. [8] investigated the effect of large density difference on film cooling effectiveness. It has been found that the density ratio has a strong effect on the film effectiveness for injection through holes. It was found that the center-line effectiveness increases somewhat with a decreasing ratio of boundary layer thickness to injection tube diameter. Ito et al. [9] used a mass transfer technique to measure the local film-cooling produced by a row of jets on a gas turbine blade. Curvature effect studied by them reveals that the lower momentum flux ratio produces higher effectiveness compared to that of flat wall. Higher momentum flux ratio lowers the effectiveness on the convex wall.

Sinha et al. [10] studied film cooling effectiveness using a row of inclined holes that injected cryogenically cooled air across a flat, adiabatic test plate. Cho and Goldstein [11] studied the Heat, mass transfer and film cooling effectiveness with injection through discrete holes. The heat, mass transfer coefficient and the film cooling effectiveness were obtained from separate tests using pure air and naphthalene-saturated vapor injected through circular holes into a cross flow of air. Abhari [12] studied the impact of rotor-stator interaction on surface heat transfer of film cooled turbine blades. Wilfert and Fottner [13] studied the aerodynamic mixing effect of discrete cooling jets with mainstream flow on a highly loaded turbine blade. Ligrani and Lee [14] presented the experimental results which described the development and structure of flow downstream of a single row of holes with compound angle orientations producing film cooling at high blowing ratios. This film cooling configuration is important because similar arrangements are frequently employed on the first stage of rotating blades of operating gas turbine engines. Abuaf et al. [15] studied the heat transfer and film cooling effectiveness in a linear airfoil cascade. Thole et al. [16] studied the complex flow of jets issuing into a crossflow from an inclined hole that has a short length-to-diameter ratio. In particular, this experimental study focused on the effect of different velocities in a coflowing channel at the cooling hole entrance. Gritsch et al. [17] tested the discharge coefficients of three film-cooling hole geometries (cylindrical and two fan-shaped holes) and showed that the discharge coefficient of both fan-shaped holes was higher than the cylindrical hole. Kercher [18] collectively described the continual emerging of numerical film cooling as a viable design tool for high temperature components. Hyams and Leylek [19] studied the physics of the film cooling process for shaped, streamwise-injected, inclined jets for different blowing ratios, density ratios, and length-to-diameter ratios. Detailed flow field results as well as surface phenomena involving adiabatic film effectiveness (η) and heat transfer coefficient (h) are presented.

For the first time in the open literature, the combined effects of compound-angle injection and hole shaping are isolated and the dominant mechanisms are examined by Brittingham and Leylek [20]. McGovern and Leylek [21] implemented a novel vorticity and momentum based approach to document how the symmetric, counterrotating vortex structure, typically found in the cross flow region in streamwise injection cases, becomes asymmetric with increasing compound angle. Lakehal et al. [22] studied the three-dimensional flow and heat transfer calculations of film cooling at the leading edge of a symmetrical turbine blade model. Cho et al. [24] investigated the local heat, mass transfer characteristics and local film cooling effectiveness around a conical-shaped film cooling hole with compound angle orientations. The effect of discrete delta- or triangular-shaped tabs with different orientations on the film cooling performance from a row of cylindrical holes was investigated by Nasir et al. [26]. Saumweber and Schulz [27]

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studied the effect of interaction of film cooling rows. Kim et al. [28] investigated the flow characteristics of the turbine blade leading edge film cooling using five different cylindrical body models with various injection holes, which are a baseline cylindrical hole, two laidback (spanwise-diffused) holes, and two tear-drop shaped (spanwise- and streamwise-diffused) holes, respectively. Effects that two different compound-angle film-hole configurations have on film cooling effectiveness are investigated by Taslim and Khanicheh [30]. Thole and Knost [31] studied the heat transfer and film-cooling for the endwall of a first stage turbine vane. Yuen and Martinez [32] studied the film cooling effectiveness experimentally on rows of cylindrical holes with streamwise angles of 30o, 60o and 90o, in a flat plate test facility with a zero pressure gradient. Miao and Ching [33] carried out the numerical simulation of film-cooling concave plate as coolant jet passes through two rows of holes with various orientations of coolant flow. This investigation considered two coolant flow orientations: (1) the coolant jets were ejected from a straight-blow supply plenum, so that the coolant supply plane is parallel to the entrance plane of the coolant jets; (2) the coolant jets were ejected from the cross-blow supply plenum so that the coolant supply plane was normal to the entrance plane of the coolant jets. Detailed analyses of computational simulations with comparisons to experimental data were performed to identify and explain the dominant flow mechanisms responsible for film cooling performance with compound angle injection of 45, 60, and 90 deg. Falcoz et al.[34] showed an experimental study of different showerhead cooling geometries on a blunt body. For these tests, TLC (thermochromatic liquid crystals) had been used for measuring the film cooling performance and the heat transfer. Detailed experimental results for the aerodynamics, film cooling effectiveness and the heat transfer enhancement are presented for different film cooling geometries. Schobeiri et al. [35] studied the film cooling on a gas turbine blade pressure side and suction side with axial shaped holes. Antivortex concept of holes for film cooling are studied by Dhungel and Ekkad [36]. Detailed film cooling measurements are presented by Lu et al. [37] on a turbine blade leading edge model with three rows of showerhead holes. Film cooling adiabatic effectiveness for axial and compound angle holes on the suction side of a simulated turbine vane was investigated by Waye and Bogard [38]. Kusterer et al [40] introduced a new cooling technology, which reaches high film-cooling effectiveness as a result of a well-designed cooling hole arrangement for interaction of two neighboring cooling jets (double-jet film cooling (DJFC) Heidmann and Ekkad [41] studied the "Novel Antivortex" turbine film cooling hole concept. This "antivortex" design is unique in that it requires only easily machinable round holes, unlike shaped film-cooling holes and other advanced concepts. The hole design is intended to counteract the detrimental vorticity associated with standard circular cross-section film-cooling holes. Rozati and Tafti [42] studied leading edge film cooling with large eddy simulation (LES).

Overall, shaped holes provide better film cooling effectiveness compare with cylindrical holes. Compound angle injection improves film cooling effectiveness for higher blowing ratios. It has less jet penetration into the mainstream. Increased exit area and lateral spreading result in a better cooling performance. Antivortex hole concept introduced by Heidmann and Ekkad [41] is an alternate to the shaped holes where the presence of sibling hole counteract the effect of vortex created by the main hole and therefore, increases the film cooling effectiveness. The summary of the hole shape and arrangement is given in Table 1. The percentage improvement in film cooling effectiveness of different shapes with respect to a cylindrical hole case is shown in Table 2.

Traditionally, designers used empirical correlations to estimate film effectiveness. With ever increasing computing capability, Computational Fluid Dynamics (CFD) has become a tool to aid in the design. The primary objective of this study is utilize CFD as a design tool for developing film hole geometries with enhanced effectiveness.

Table 1 Summary of hole shapes.

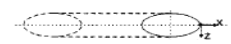
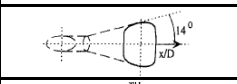
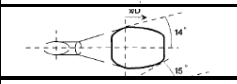
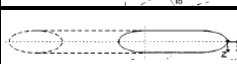
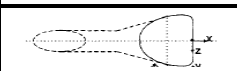
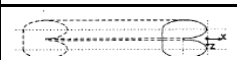
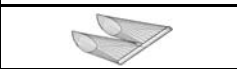
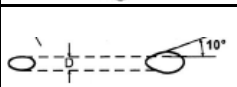
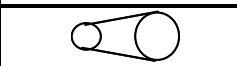
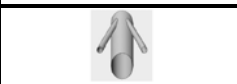
Shape 1 – Cylindrical Gritsch, Schulz and Wittig [17]	
Shape 2 – Fan shape Gritsch, Schulz and Wittig [17]	
Shape 3 – Laid back fan shape Gritsch, Schulz and Wittig [17]	
Shape 4 – Forward diffused Hyams and Leylek [19]	
Shape 5 – Lateral-Diffused Hyams and Leylek [19]	
Shape 6 – Cusp shaped Hyams and Leylek [19]	
Shape 7 – CONSOLE Sargison [23]	
Shape 8 – Forward and lateral diffusion Yu et al. [25]	
Shape 9 – Tabs Nasir et al. [26]	
Shape 10 – Conical Taslim and Ugarte [29]	
Shape 12 – Crescent shaped hole Lu [39]	
Shape11 – Cylindrical main hole and side hole Heidmann and Ekkad [41]	

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Table 2 Improvements in film cooling effectiveness

Hole Shape	% Improvement in η	References
Fan shape	10 – 40	Brittingham et al. [20], Sargison [23], Yu et al [25], Kim [28], Miao et al. [33], Lu et al. [37]
Console	20	Sargison [23]
Conical	15	Falcoz et al. [34]
Sister hole	15 – 23	Dhungel [36]
Compound angle	4 - 10	Waye et al. [38]
Trench shape	15 – 20	Baheri et al.[43]

CFD Code Validation

Prior to utilize CFD as a design tool, its film effectiveness predictive capability was assessed by modeling the film cooling case presented in the experimental work of Sinha et al. [10]. The study was conducted on a flat plate with cylindrical holes fed by a plenum and angled 35° in the mainstream direction (Figure 1). Two blowing ratios (M) 0.5 and 1.0 were modeled with a density ratio (DR) of 2.0.

The commercially available CFD code ANSYS-CFX-11.0 has been used to perform the flow simulations. Turbulence was modeled with the k-ε model with scalable wall function. The mesh, which was created in ANSYS-ICEM-CFD-11.0, consisted of a structured grid in the main gas path and a tetrahedral mesh in the cooling hole (Figure 2). The film hole mesh was attached to the main path via an non-conformal grid interface. This strategy of separating the film hole geometry from the other domains is to evaluate multiple hole geometries quickly. Inlet boundary conditions for the main flow consisted of velocity of 20 m/s and a total temperature of 300K. The main flow exit was set to a static pressure of 1 atm. The coolant had a temperature of 150K and the mass flow rate was specified to satisfy the blowing ratios. All the walls were set adiabatic heat transfer and no-slip conditions.

The performance of cooling can be expressed in terms of adiabatic film cooling effectiveness

$$\eta = (T_w - T_m)/ (T_c - T_m)$$

where, T_c is the stagnation temperature of the cooling fluid near entry; and T_w is the adiabatic wall temperature. T_m is the stagnation temperature of the mainstream air. Figure 3 shows the centerline film effectiveness comparison between CFD and experimental data. There is a good agreement with experimental data for X/D > 10. Near the hole exit for the lower blowing ratio film effectiveness is under predicted. For M=1, there is good agreement for X/D < 5. However, CFD does not capture the effectiveness increase for 5 < X/D < 15 (for M=1). CFD shows that film effectiveness for M=0.5 decays at a faster rate than M=1.0 (for X/D>15). Film blow-off is also captured for M=1.0 denoted by sudden effectiveness fall near the hole exit. Furthermore, CFD demonstrates that away from the hole exit, film effectiveness is superior for the higher blowing ratio. The centerline effectiveness (Fig. 3) as well as the lateral averaged effectiveness (Fig. 4) compares favorably with the test data when compared to the previous study of Ely & Jubran [44]. In general, CFD depicts film behavior adequately. Therefore, the CFD model as represented here could be used to quickly develop, evaluate and optimize new film hole geometries.

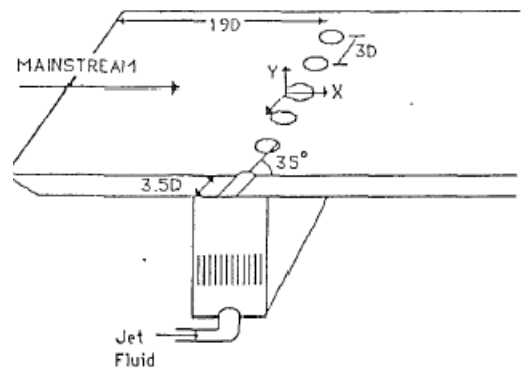

Figure 1. Experimental Configuration from Sinha, et al. [10].

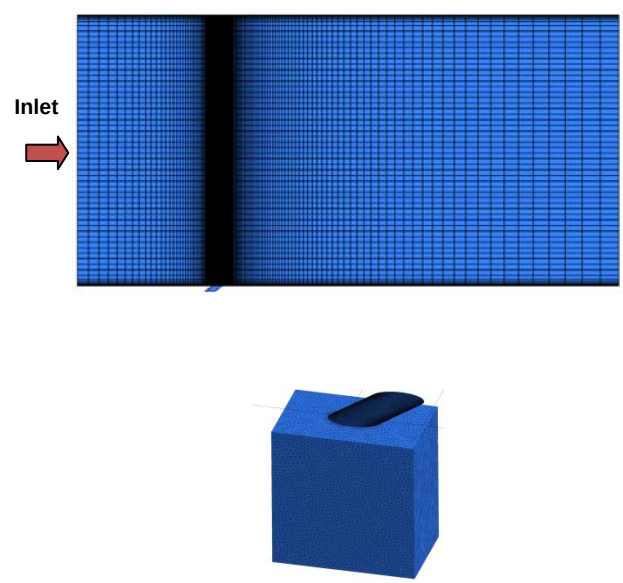

Figure 2. CFD Mesh.

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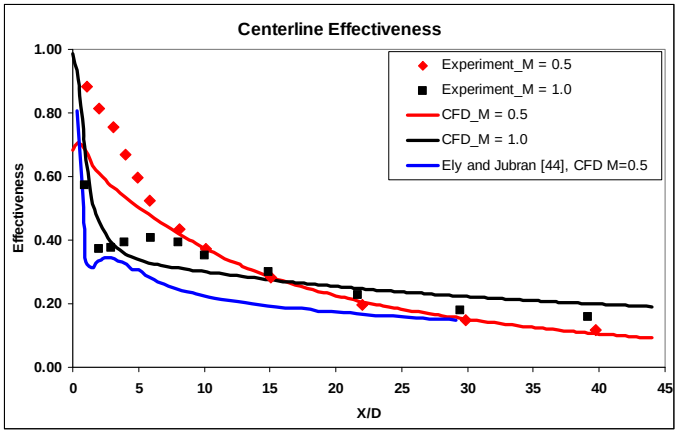

Figure 3. Centerline Effectiveness for Case with Plenum.

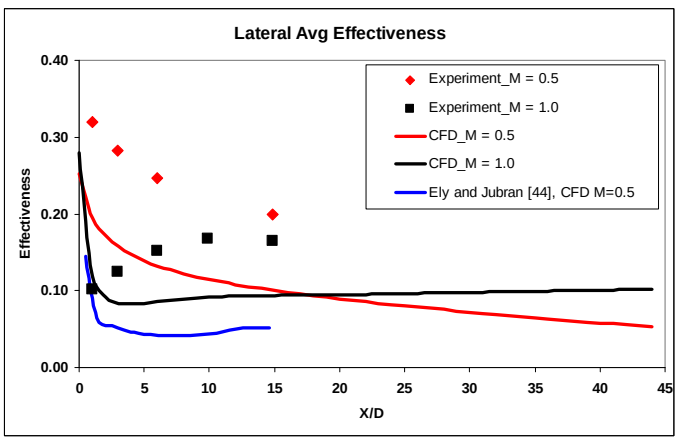

Figure 4. Lateral Average Effectiveness for Case with Plenum.

OPTIMIZATION OF FAN SHAPED HOLE

It is known that a film cooling hole with a diffusion in its passage leads to a higher effectiveness Yu et al. [25]. In order to incorporate this diffusion effect on the coolant flow it is necessary to provide a forward diffused or lateral diffused angle or a combination of both at the exit of the film cooling hole (Fig. 5). These angles need to be optimized for achieving higher effectiveness value. A Design of Experiment (DoE) was conducted for various values of Lateral and Forward angle combinations shown in Table 3.

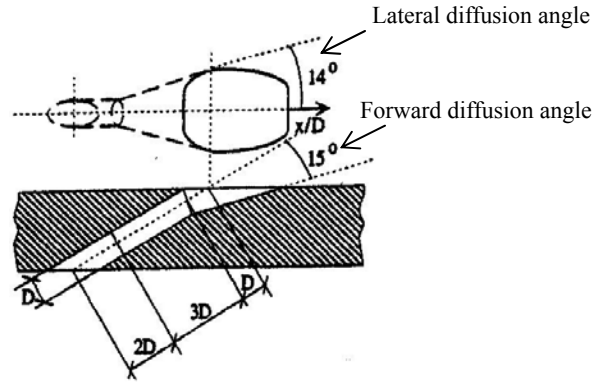

Figure 5. Fanshaped hole with lateral and forward angled diffusion.

Table 3 Fan shaped hole configurations.

Cases	Lateral angle (deg)	Forward angle (deg)
1	10	10
2	14	10
3	10	14
4	14	14
5	12	12
6	0	0

These cases were evaluated for three different blowing ratios of 0.5, 1.0 and 1.5 in a flat plate arrangement. Although the analyses were carried out with the flat plate, the mainstream Mach number was increased from 0.2 to 0.9 to simulate a typical turbine accelerating flow case. This was done by including a curved wall on the top of the domain (Figure 6).

The grid generation involved a hybrid meshing of the domain where a tetrahedral meshing is carried out for hole domain and hexahedral meshing is done for the mainstream channel.

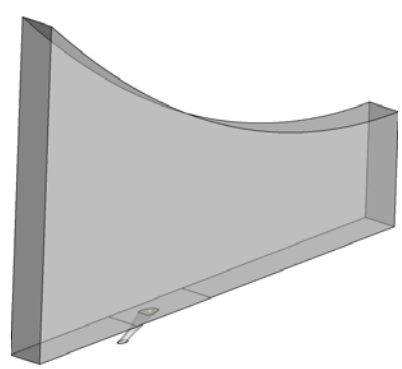

Figure 6. Computational domain.

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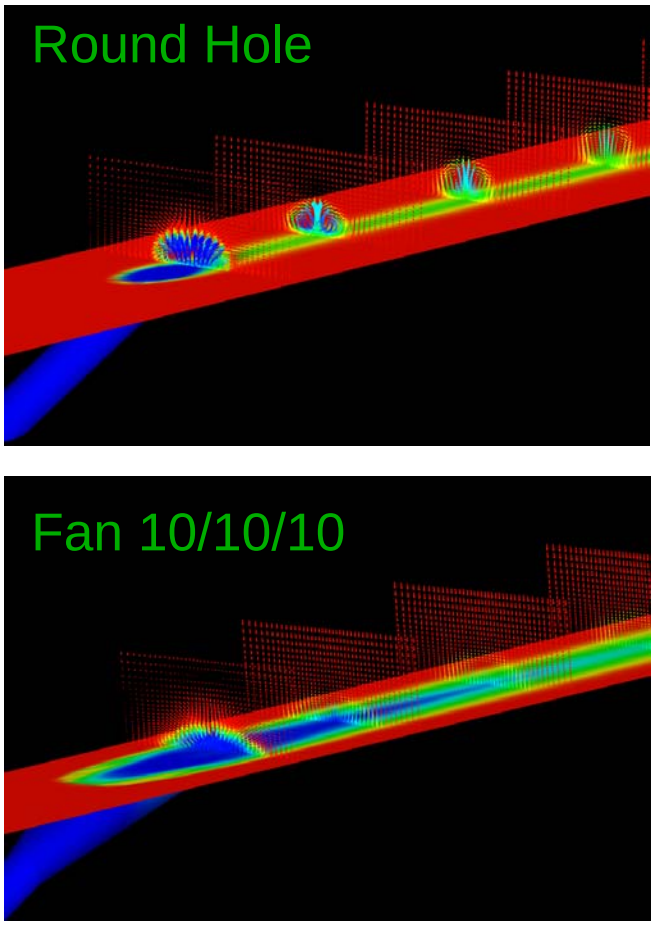
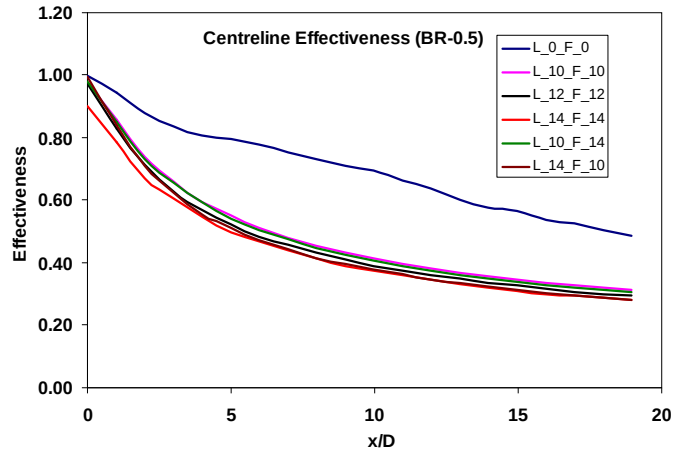


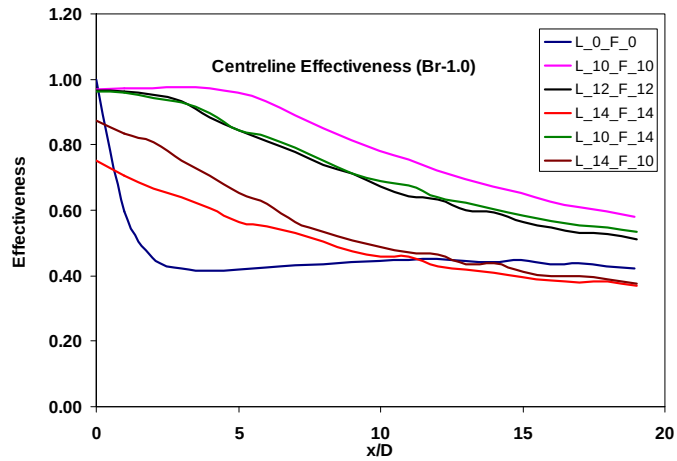
Figure 7. Flow field from cylindrical and fan shaped holes for M=1.

Figure 7 shows the flow field on the normal plane at different x/d locations to visualize the coolant and mainstream interactions. The flow field from the round holes shows the effect of kidney vortices in mixing the coolant flow with the main flow. The fan shaped hole has a more lateral spread and weaker vortices which will improve the coolant performance.

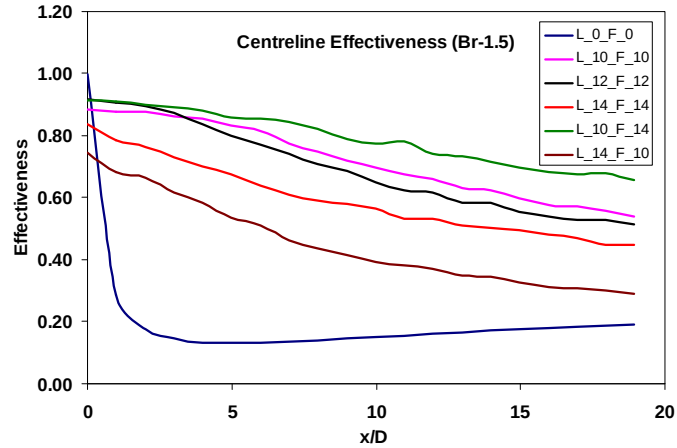
Centerline effectiveness, shown in Figure 8, indicates the amount of coolant flow adhering to the blade surface to protect it. Effectiveness is compared among the different fan shaped holes that were analyzed. Cylindrical hole had the highest value of centerline effectiveness at low blowing ratio of 0.5. At higher blowing ratios, the performance of cylindrical hole deteriorates due to the increase in the exit momentum of the coolant fluid. Cylindrical hole also has lower lateral effectiveness due to the lack of lateral spreading of the cooling flows. Fan shaped holes give higher effectiveness at higher blowing ratios due to the diffusion at the exit of the hole. Forward and lateral diffusion angles play an important role in determining the cooling performance. Forward angle reduces the flow separation at the hole exit which helps to retain the coolant flow closer to the blade surface. With increase in the blowing ratio, it is found that the cases with lateral and forward diffused angle 10/10/10 and 10/10/14 give higher value of centerline effectiveness when compared to the other configurations. For higher blowing ratio of 1.5, case 10/10/14 gives higher centerline effectiveness than 10/10/10 due to the forward angle diffusion.



(a) Blowing ratio = 0.5



(b) Blowing ratio = 1.0



(c) Blowing ratio = 1.5

Figure 8. Centerline Effectiveness for Fan Shaped Holes.

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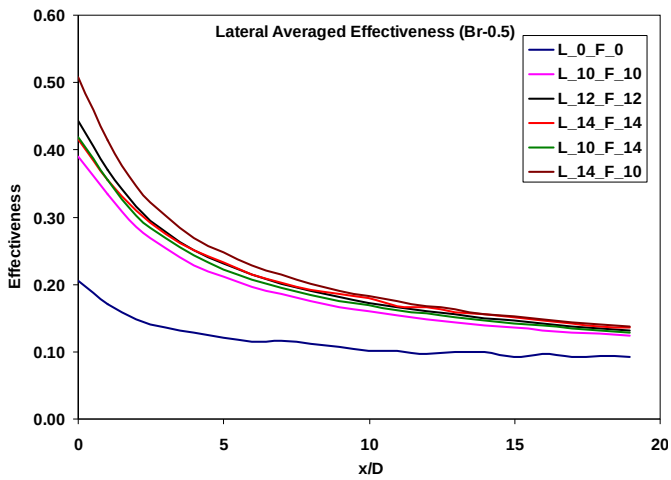
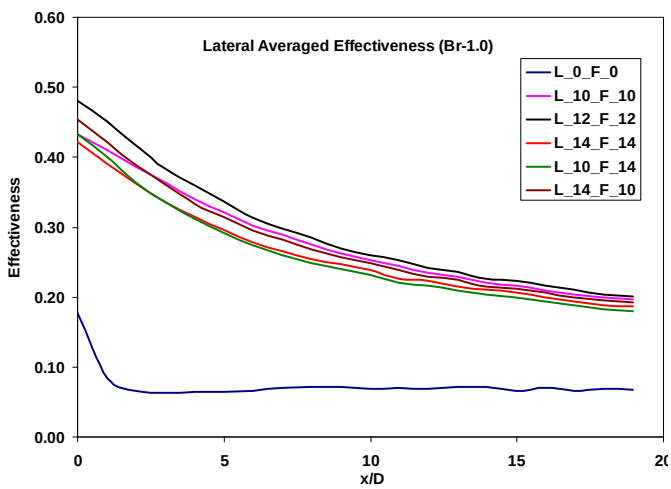
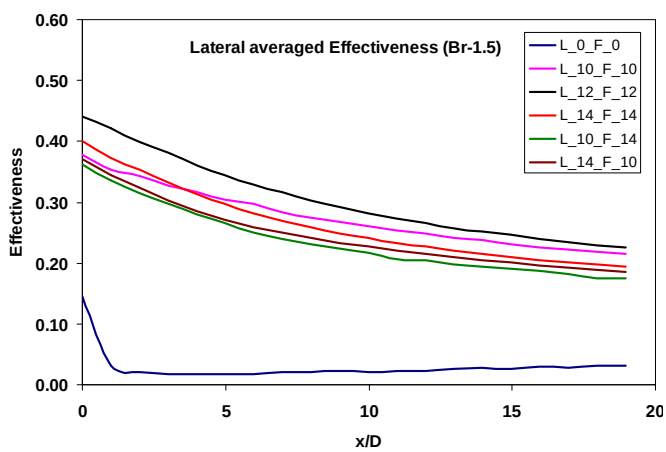


(a) Blowing ratio = 0.5



(b) Blowing ratio = 1.0



(c) Blowing ratio = 1.5

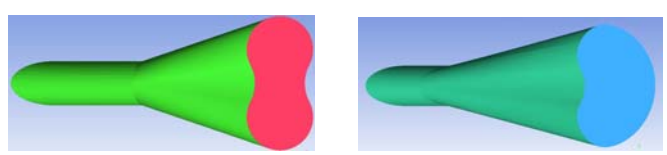
Figure 9. Lateral averaged Effectiveness for Fan Shaped Holes.

Lateral averaged effectiveness distribution for different cases is shown in Fig 9. The performance of cylindrical hole is very poor when compared to fan shaped holes for all blowing ratios. For the high blowing ratios increase in the lateral diffusion angle more than 12 deg, the lateral averaged effectiveness decreases due to the flow separation inside the hole passage. The flow deviates towards one side of the cooling hole path. This results in an asymmetric flow pattern at the exit of the hole which in turn leads to poor lateral spread of the cooling flow over the blade surface. The case 10/10/10 and 12/12/12 has higher lateral averaged effectiveness. Overall, case 10/10/10 gives the highest value of both centerline effectiveness and lateral averaged effectiveness, which is taken as a baseline for comparison subsequently.

NOVEL FILM HOLE GEOMETRIES

In pursuit of enhancing the film cooling performance further than that of the fan shaped hole it was decided to find new shapes which reduce the effect of vortices and cooling jet penetration into the mainstream, which in turn reduce effectiveness.

Since the kidney vortices play an important role in mixing, any method to reduce the strength of the kidney vortex could be expected to improve the effectiveness. It was therefore decided to try a shape similar to a dumb-bell which could be expected to reduce the strength of kidney vortex by virtue of its shape which splits the circular exit into two smaller circles (Fig. 10.a). As would be seen later, the centerline effectiveness is lower (compared to that of fan shaped hole) for the dumbbell shape and a correction was hence applied and the resulting shape is that of a bean (Fig. 10. b).



(a) Dumbbell shape (b) Bean shape

Figure 10. Novel film hole geometries.

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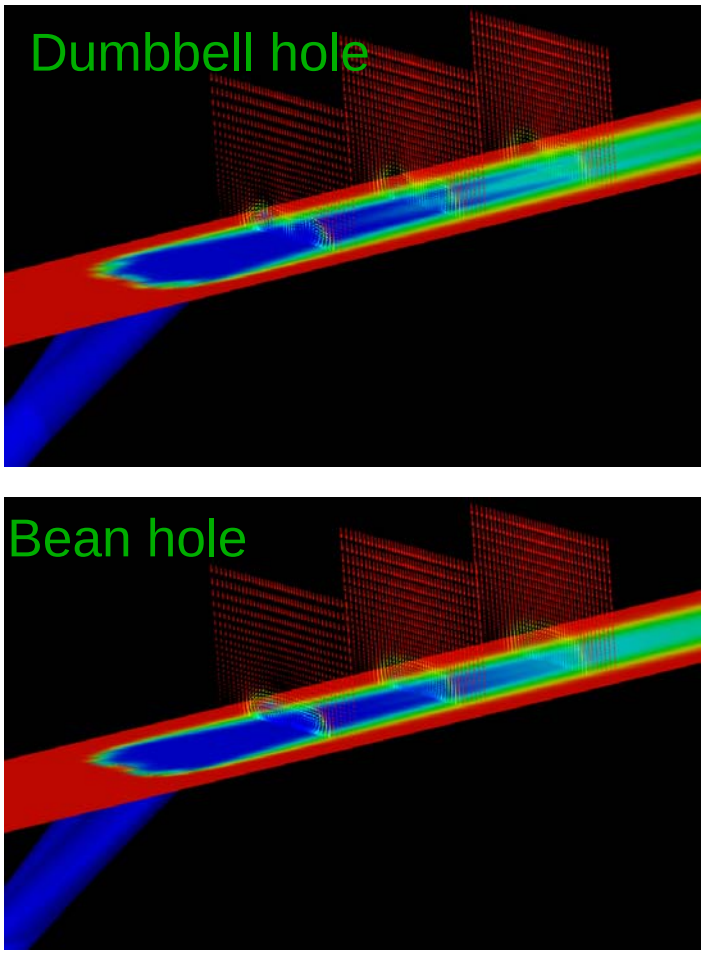

Figure 11. Flow field from dumbbell and bean shaped holes for M=1.

In addition it was thought that the spreading of the hole in the lateral direction (for a given area ratio) would enhance lateral spread of the coolant. The results from the CFD simulations confirm this trend (Fig. 11 & 12).

In the case of dumb-bell and bean shapes, the exiting jet on the lateral side helps to protect the center coolant flow from the entrainment of the mainstream thereby increases the effectiveness. This new shape will minimize the coolant liftoff by reducing the vortex interaction with the mainstream (Fig .12). In particular for the bean shaped hole the convexity on the trailing edge provides more flow on the centerline which provides better film cooling on the blade surface.

Initial analysis is carried out with the same area ratio between exit and inlet, except that the exit shape is changed to dumbbell or bean instead of a cylindrical. The new shapes allow the flow to diffuse along the hole passage due to increase in the lateral dimension. This change in the coolant flow passage helps in increasing the lateral spread of the coolant flow on the blade surface. Before comparing with the fan shaped hole, optimization study is carried out for the new shaped hole with respect to the area ratio. It is found that the area ratio of 2.5 for both dumbbell and bean shape had the highest value of effectiveness.

Effectiveness contours for optimized fan shape, bean shaped and dumbbell shaped holes are shown in Fig. 12 for three different blowing ratios 0.5, 1.0 and 1.5. It can be seen that the lateral spread in the case of dumbbell and bean shaped hole was more when compared to the fan shaped hole. At higher blowing ratio there are asymmetries in the flow pattern at the exit of the hole in the case of optimized fan shaped hole (Fig 12.c). Asymmetry is not predominant in the case of dumbbell and bean shaped holes and can be seen to have a widespread coolant flow distribution on the blade surface resulting in effective film cooling.

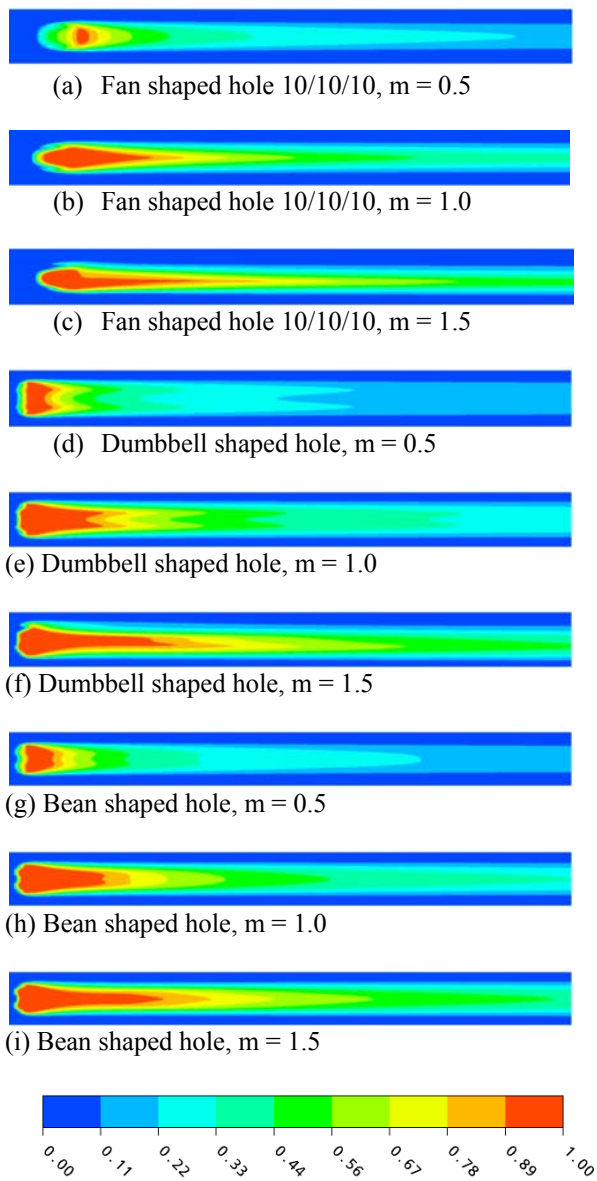

Figure 12. Film Effectiveness Contours for optimized fan, dumbbell and bean shaped holes

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Comparison of centerline and the lateral averaged effectiveness among the fan shape, dumbbell and bean shape are shown in Fig. 13 & 14. The Centerline effectiveness values are extracted at different x/d locations starting from the hole exit x/d=0 to x/d=45. The average centerline effectiveness is calculated from the averaging of the extracted values. Graph is plotted between different area ratio Vs averaged effectiveness value, the dashed line shows the optimum Fan shaped hole effectiveness value. Solid line indicates the effectiveness value of the new shape with different area ratio. From the centerline effectiveness plot for dumbbell shape it is found that the average effectiveness is lesser than that of the fan shaped hole except for the blowing ratio 1.5. At blowing ratio 1.5 it is found that the flow pattern is asymmetrical at the exit for the fanshaped hole (as seen in Fig 12.c) that leads to lower centerline effectiveness for the fanshaped hole.

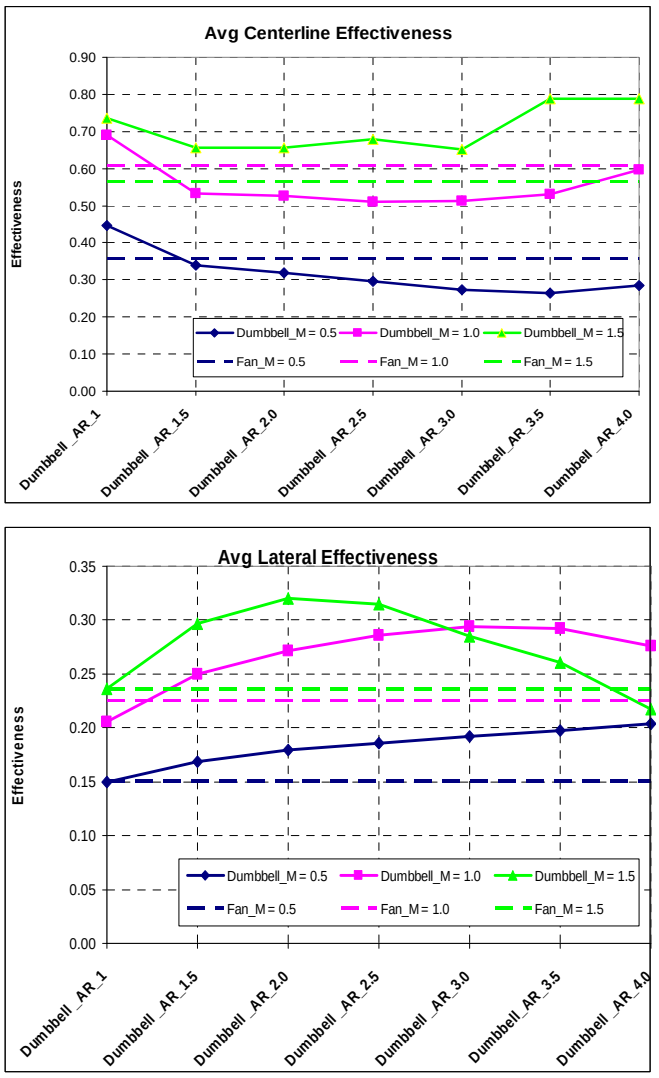

Figure 13. Overall Effectiveness comparisons for optimized fan and dumbbell shaped holes.

In the case of bean shaped hole the averaged centerline effectiveness is higher when compared with fan shaped hole. But the effectiveness decreases for higher area ratio (AR > 2.0). This is due to the fact that at higher area ratio the chances for the flow to separate inside the coolant flow passage is more. In general lateral spread of the coolant flow is more important than the centerline effectiveness.

In the case of lateral averaged effectiveness for dumbbell shape it is observed that the effectiveness value increases with the area ratio for blowing ratio 0.5 and 1.0. The values are higher when compared to the fan shaped hole. By comparing the effectiveness value for all the blowing ratios it is found that the area ratio of around 2.5 gives the higher value of effectiveness. Even in the case of Bean shaped hole the effectiveness value increases with increase in the area ratio for Br. 0.5 and 1.0. For a blowing ratio of 1.5 it is seen that the effectiveness value decreases beyond area ratio 2.5.

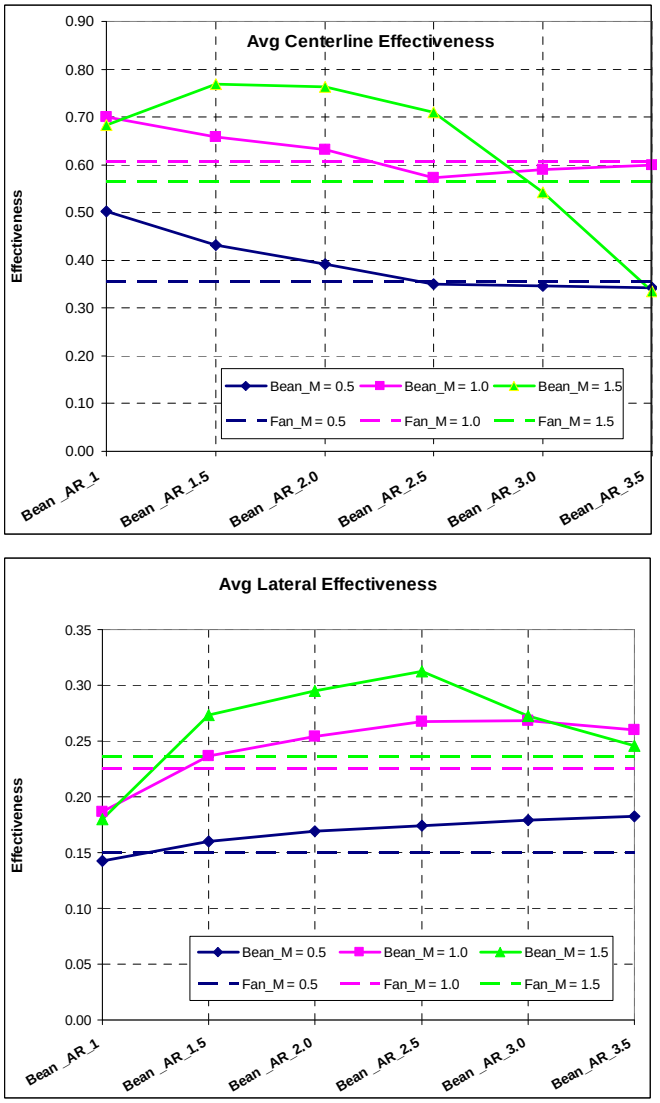

Figure 14. Overall Effectiveness for optimized fan and bean shaped holes.

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CONCLUSIONS

Film cooling effectiveness for round hole, different fan shaped holes, and two novel shaped holes geometries were analyzed using CFD for flow over a flat plate under favorable pressure gradient. Round hole results were compared with experimental data to validate the CFD processes. The DOE study revealed that the fan shape configuration with lateral and forward diffused angle of 10° gives the highest effectiveness among all fan shaped holes.

Two new shapes were developed to further enhance the film cooling effectiveness over than the fan shaped holes. The new shapes were further optimized. An area ratio of 2.5 has the highest value of film cooling effectiveness. Overall improvement of approximately 33% in average lateral effectiveness is observed with the new shaped holes over the fan shaped film cooling holes.

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